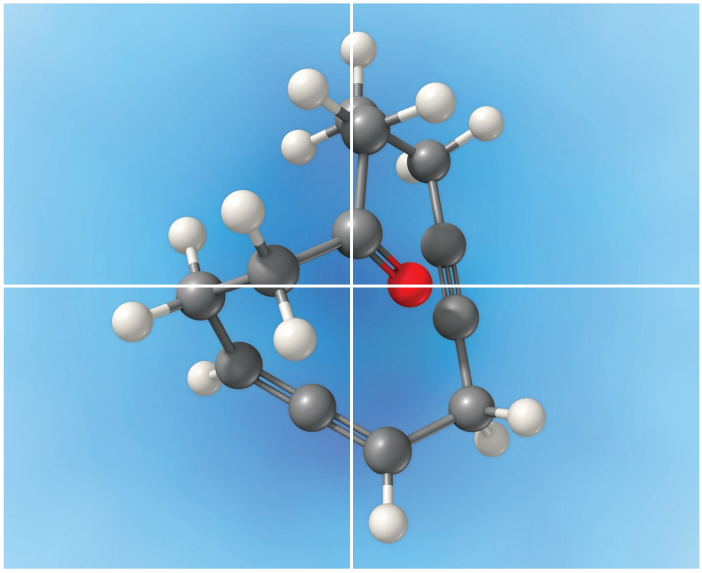
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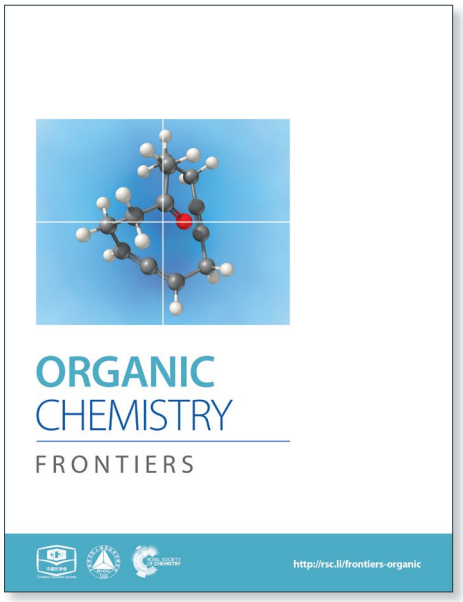
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ARTICLE

Suzuki coupling for preparation of allenes-ligand effect and chirality transfer

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

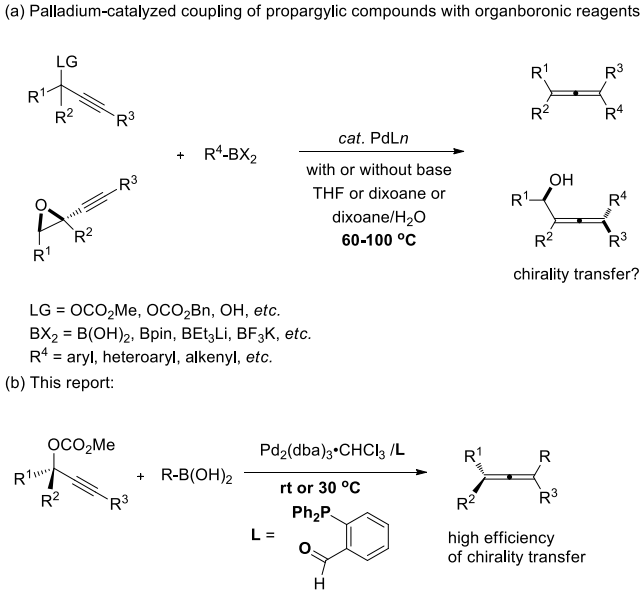
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Hongwen Luo[a], Yihua Yu[b] and Shengming Ma[c,d]

A general and efficient palladium-catalyzed coupling of propargylic carbonates with organoboronic acids has been developed by identifying a commercially available semi-labile bidentated *o*-(diphenylphosphino)benzaldehyde as the effective ligand affording allenes under a set of very mild reaction conditions. Many functional groups are tolerated in this reaction, and high chirality transfer has also been realized in the presence of water leading to a very rapid reaction for the synthesis of easily recemized 1,3-diarylallenes. A rationale has been proposed for the role of the ligand.

Introduction

Allenes are present in many natural products and have been recognized as very useful building blocks in organic synthesis nowadays.[1,2] Consequently, extensive studies have been focused on efficient methods for the synthesis of different types of allenes.[2j,3] Palladium-catalyzed coupling of propargylic compounds with organometallic reagents,[4] such as Grignard reagents,[5] organoaluminum reagents,[6] organoindium reagents,[7] organozinc reagents,[5b,6a,8] organocopper reagents,[5b] organosilver reagents,[5b] organotin reagents,[6a,9] and organosilicone reagents,[10] is one of the most useful and efficient methodology to synthesize allenes. However, some organometallic reagents are unstable, expensive, and toxic. Although related reports on utilizing these easily available mild organoboronic compounds have been reported, a temperature above 60 °C is required in most of those studies, which may be troublesome for the easy racemizing nature of allenes (Scheme 1, a):[11,12] Thus, report on the efficient chirality transfer using optically active propargylic derivatives for such allene syntheses are very limited. Molander and Yoshida independently reported that a decent level of chirality transfer in the coupling reaction of propargylic phosphates with potassium trifluoroborates[12a] or terminal propargylic carbonates with aryl boronic acids,[11f] however, in majority cases, serious racemization or lower efficiency of chirality transfer was observed. The challenges are the ready racemization of propargylic/allenylic palladium intermediates through the σ-π-σ isomerization process[13] and the sensitivity of allenes (see also Scheme 4). With our goal of developing asymmetric synthesis of allenes, we wish to develop such a coupling reaction using commercially available boronic acids under ambient conditions (Scheme 1, b).



Scheme 1 Pd-catalyzed coupling protocols of propargylic derivatives with organoboronic compounds.

As we know, the formyl group (CH=O) has been explored for its coordination ability with transition metals, such as Ru, Rh, Re, Cu, and Pd.[14] However, to the best of our knowledge, only very few studies employed those complexes as catalysts have been reported,[15] probably due to the high reactivity or sensitivity of aldehyde functionality under different reaction conditions. In this report, we wish to disclose that the commercially available *o*-(diphenylphosphino)benzaldehyde has been identified as the effective ligand for the allene synthesis under very mild reaction conditions with a very wide scope both for arylboronic acids and propargylic carbonates. In

[a] Shanghai Key Laboratory of Green Chemistry and Chemical Process, Department of Chemistry, East China Normal University, 3663 North Zhongshan Lu, Shanghai 200062, P. R. China
[b] Department of Physics, East China Normal University, 3663 North Zhongshan Lu, Shanghai 200062, P. R. China
[c] Department of Chemistry, Fudan University, 220 Handan Lu, Shanghai 200433, P. R. China
[d] State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China

† Electronic Supplementary Information (ESI) available: Experimental procedure, spectroscopic data, and the ^{1}H, ^{13}C, ^{19}F NMR spectra of all the products. See DOI: 10.1039/x0xx00000x

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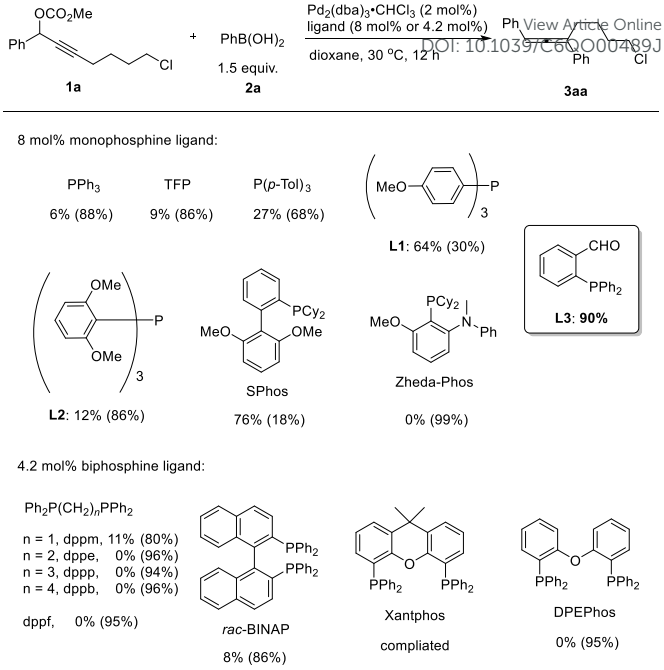
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addition, high chirality transfer has also been realized with the help of water.

Results and Discussion

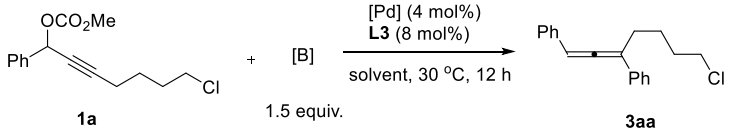
In our initial studies, 7-chloro-1-phenylhepta-2-yn-1-yl methyl carbonate **1a** and phenylboronic acid **2a** were used as the prototypical substrates for optimization of reaction conditions. When a mixture of **1a** and **2a** was treated with $Pd_2(dba)_3{\cdot}CHCl_3$ (2 mol%) and PPh_3 (8 mol%) in dioxane at 30 °C for 12 h, only 6% yield of the corresponding allene product **3aa** was obtained (Scheme 2). Although electron-rich ligand **L1** gave 64% yield of product **3aa**, other monodentate ligands, such as TFP and P(*p*-Tol)$_3$ failed to promote the reaction smoothly. Then, we studied the effect of bidentate ligands. SPhos with OMe as the flexible coordination site provided an acceptable yield of product **3aa** with 18% of recovery, however, ligand **L2** and Zheda-Phos failed in the reaction. To our surprise, the commercially available bidentate monophosphine, *o*-(diphenylphosphino)benzaldehyde (**L3**),[16] which has an interesting highly reactive formyl unit in the framework, performed efficiently offering the highest yield of 90%. Control experiments showed that commonly available bisphosphine ligand, such as $Ph_2P(CH_2)_nPPh_2$ (*n* = 1-4), dppf, *rac*-BINAP, Xantphos, and DPEPhos, afford a result of less than 11% yield, indicating the unique role of the formyl group.

After identifying **L3** as the efficient ligand, other parameters were then investigated (Table 1). Although $Pd(OAc)_2$ afforded **3aa** with 77% yield, $PdCl_2$ and $[Pd(allyl)Cl]_2$ were both ineffective for the present transformation, yielding only trace quantity of the targeted product (Table 1, entries 1-3). No better results were obtained in toluene, DCE, or THF (Table 1, entries 5-7). We also studied the reactivity of other boron reagents (Table 1, entries 8-10): The reaction using PhBpin instead of $PhB(OH)_2$ offered 42% yield of product, while $(PhBO)_3$ (0.5 equiv.) and $NaBPh_4$ (0.4 equiv.) afforded **3aa** in 82% and 90% yields, respectively. The reaction also process smoothly at room temperature (25 °C) with a slightly



Scheme 2 Ligand optimization of the Reaction Conditions. The yield and recovery were determined by ^{1}H NMR analysis, and the recovery of **1a** was displayed in the parentheses.

Table 1 Optimization of the Reaction Conditions[a]



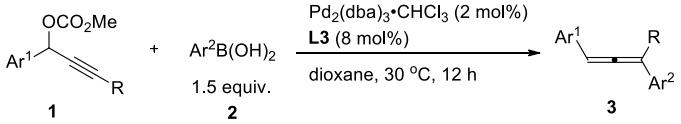
Entry	[Pd]	Solvent	[B]	Yield (%)[b]	Recovery (%)[b]
1	$PdCl_2$	dioxane	$PhB(OH)_2$	-	100
2	$Pd(OAc)_2$	dioxane	$PhB(OH)_2$	77	14
3	$[Pd(allyl)Cl]_2$	dioxane	$PhB(OH)_2$	5	91
4	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	$PhB(OH)_2$	90	-
5	$Pd_2(dba)_3{\cdot}CHCl_3$	toluene	$PhB(OH)_2$	60	30
6	$Pd_2(dba)_3{\cdot}CHCl_3$	DCE	$PhB(OH)_2$	24	57
7	$Pd_2(dba)_3{\cdot}CHCl_3$	THF	$PhB(OH)_2$	63	16
8	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	PhBpin	42	47
9	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	$(PhBO)_3$[c]	82	6
10	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	$NaBPh_4$[d]	90	-
11[e]	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	$PhB(OH)_2$	84	-
12[f]	$Pd_2(dba)_3{\cdot}CHCl_3$	dioxane	$PhB(OH)_2$	91 (80[g])	-

[a]The reaction was conducted with 0.5 mmol of **1a**, 0.75 mmol of [B], palladium source (0.02 mmol of Pd atom), and 0.04 mmol of **L3** in 3 mL of solvent for 12 h at 30 °C under Ar atmosphere. [b]Determined by ^{1}H NMR analysis with CH_2Br_2 as the internal standard. [c]0.25 mmol of $(PhBO)_3$ was uesed. [d]0.2 mmol of $NaBPh_4$ was used. [e]The reaction was conducted on 25 °C. [f]The reaction was conducted with 1 mmol of **1a** in 2 mL of dioxane. [g]Isolated yield.

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Table 2 Palladium-catalyzed coupling reaction of propargylic carbonate **1** with arylboronic acid **2**[a]



Entry	1		2	Yield of **3** (%)[b]
	Ar^1	R	Ar^2	
1	Ph	$(CH_2)_4Cl$ (**1a**)	Ph (**2a**)	80 (**3aa**)
2	Ph	$(CH_2)_4Cl$ (**1a**)	4-MeC_6H_4 (**2b**)	64 (**3ab**)
3	Ph	$(CH_2)_4Cl$ (**1a**)	3-MeC_6H_4 (**2c**)	69 (**3ac**)
4	Ph	$(CH_2)_4Cl$ (**1a**)	1-Naphthyl (**2d**)	70 (**3ad**)
5	Ph	$(CH_2)_4Cl$ (**1a**)	4-$OHCC_6H_4$ (**2e**)	71 (**3ae**)
6	Ph	$(CH_2)_4Cl$ (**1a**)	4-AcC_6H_4 (**2f**)	82 (**3af**)
7[c,d]	Ph	$(CH_2)_4Cl$ (**1a**)	4-BrC_6H_4 (**2g**)	74 (**3ag**)
8	Ph	$(CH_2)_4Cl$ (**1a**)	3-$O_2NC_6H_4$ (**2h**)	80 (**3ah**)
9	Ph	$^nC_6H_{13}$ (**1b**)	4-$EtO_2CC_6H_4$ (**2i**)	76 (**3bi**)
10	Ph	$^nC_6H_{13}$ (**1b**)	4-$MeOC_6H_4$ (**2j**)	82 (**3bj**)
11[e]	Ph	^{n}Bu (**1c**)	2-MeC_6H_4 (**2k**)	77 (**3ck**)
12	Ph	^{n}Bu (**1c**)	4-$MeOC_6H_4$ (**2j**)	80 (**3cj**)
13[f]	Ph	^{n}Bu (**1c**)	4-$MeOC_6H_4$ (**2j**)	88 (**3cj**)
14	Ph	^{n}Bu (**1c**)	3-$MeOC_6H_4$ (**2l**)	79 (**3cl**)
15[c,d]	Ph	^{n}Bu (**1c**)	2-$MeOC_6H_4$ (**2m**)	72 (**3cm**)[g]
16	Ph	$(CH_2)_2Ph$ (**1d**)	Ph (**2a**)	65 (**3da**)
17	Ph	^{i}Pr (**1e**)	4-$MeOC_6H_4$ (**2j**)	78 (**3ej**)[g]
18[d]	4-$MeO_2CC_6H_4$	^{n}Bu (**1f**)	Ph (**2a**)	75 (**3fa**)
19	4-NCC_6H_4	^{n}Bu (**1g**)	Ph (**2a**)	78 (**3ga**)[g]
20	1-Naphthyl	^{n}Bu (**1h**)	Ph (**2a**)	70 (**3ha**)
21	4-ClC_6H_4	^{n}Bu (**1i**)	4-$MeOC_6H_4$(**2j**)	84 (**3ij**)
22	4-BrC_6H_4	^{n}Bu (**1j**)	4-$MeOC_6H_4$(**2j**)	87 (**3jj**)
23	2,4-$Cl_2C_6H_3$	^{n}Bu (**1k**)	3-$MeOC_6H_4$(**2l**)	68 (**3kl**)
24	4-MeC_6H_4	^{n}Bu (**1l**)	3-$MeOC_6H_4$(**2l**)	73 (**3ll**)

[a]The reaction was conducted with 1 mmol of **1**, 1.5 mmol of **2**, 0.02 mmol of $Pd_2(dba)_3$•$CHCl_3$, and 0.08 mmol of **L3** in 2 mL of dioxane for 12 h at 30 °C under Ar atmosphere. [b]Isolated yield. [c]18 h. [d]2.5 mmol of arylboronic acid was used. [e]40 °C. [f]10 mmol of **1c** was used. [g]<5% of **1** were detected by ^{1}H NMR analysis with CH_2Br_2 as the internal standard.

lower yield (Table 1, entry 11). Finally, for consideration of practicality, a reaction on 1 mmol scale at a concentration of 0.5 M was applied as the standard to afford the product **3aa** in 80% isolated yield (Table 1, entry 12).

With the optimized conditions in hand, the substrate scope and generality was investigated and the results for the synthesis of 1,3-diphenylallenes are summarized in Table 2. The $Pd_2(dba)_3$·$CHCl_3$/**L3** combination demonstrated high efficiency and various functional groups are compatible: In terms of the substituent effects of arylboronic acids, Ar^2 may bear the electron-donating substituents, such as 2/3/4-methyl, 2/3/4-methoxy, and naphthyl groups affording allene products **3ab**-**3ad**, **3bj**-**3cm**, **3ej**, and **3ij**-**3ll** with good yields (Table 2, entries 2-4, 10-15, 17, and 21-24); it should be noted that more arylboronic acid or higher temperature (40 °C) were required to have a complete conversion with the more hindered 2-methyl and 2-methoxy substituted substrates **2k** and **2m**, probably due to the steric effect (Table 2, entries 11 and 15). The electron withdrawing and useful functional groups, such as -CHO, -Ac, -Br, $-NO_2$, and $-CO_2Et$ were also tolerated under this set of mild reaction conditions, which make these allene products very useful for further transformation (Table 2, entries 5-9). The reaction of propargylic carbonates bearing electron-donating substituents, such as 4-methyl and naphthyl, or electron-withdrawing substituents, such as 4-cyano and 4-ester, afforded the corresponding allenes **3fa**-**ha**, **3ll** in high yields (Table 2, entries 18-20 and 24). In addition, the substrates with a halogen group (Cl, Br) in Ar^1 also performed well, showing the potential of this reaction (Table 2, entries 21-23). For R group in

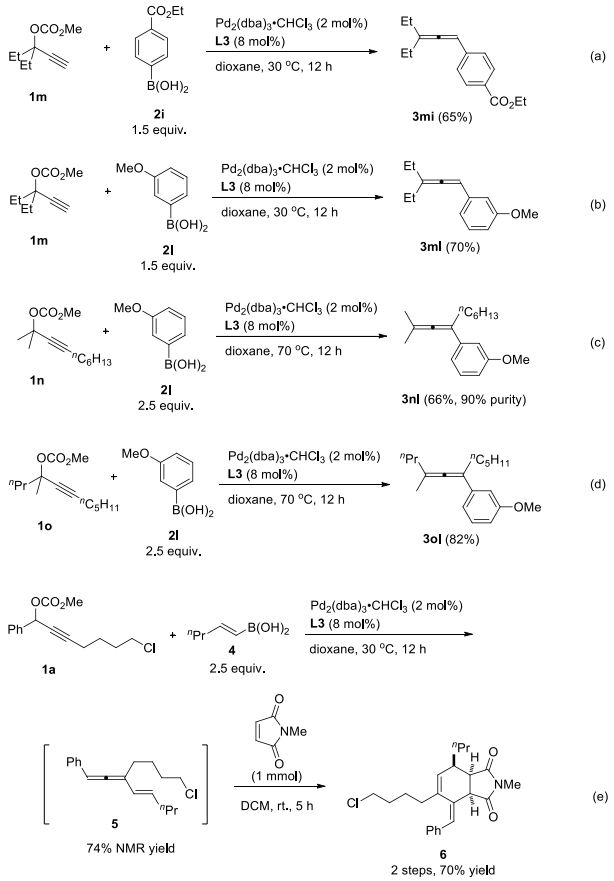
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propargylic carbonates, simple alkyl groups ($(CH_2)_2Ph$, $^nC_6H_{13}$, ^{n}Bu) and more sterically hindered alkyl group (^{i}Pr) were also suitable (Table 2, entries 9-23). Gram-reaction of **1c** with **2j** proceeds smoothly at 30 °C to give allene product **3cj** in 88% isolated yield (Table 2, entrie 13).

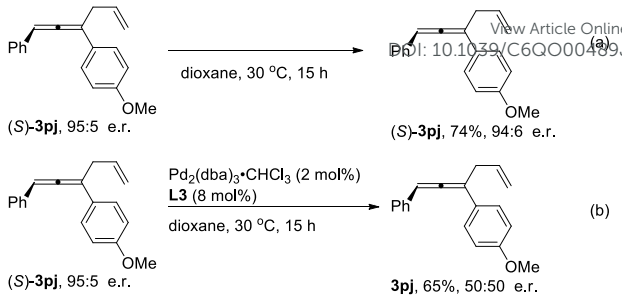
Tertiary propargylic carbonate is another type of compatible substrates for the synthesis of the tri- or even tetra-substituted allenes (Scheme 3 (a-d)). Moreover, alkenyl boronic acid is also suitable for this protocol, and easily transferred to interesting bicyclic compound **6** *in situ* due to the instability and high activity of triene **5** (Scheme 3 (e)).[17,18]



Scheme 3 Synthesis of tri- or tetra-substituted allenes from other type of propargylic carbonates or alkenyl boronic acids

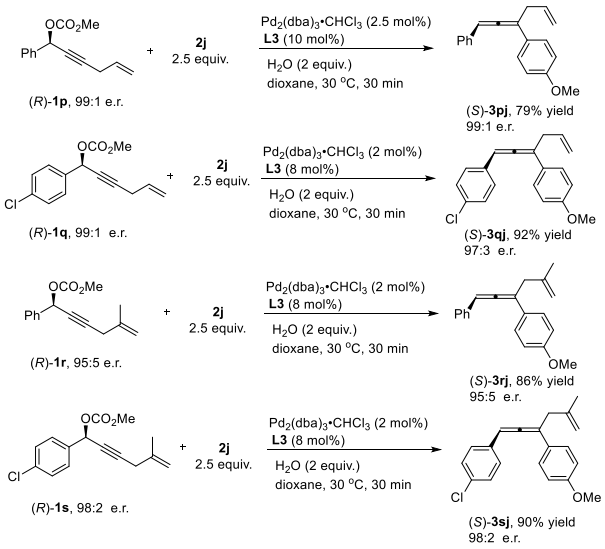
The reaction of propargylic carbonates bearing phenyl (**1a**), electron-withdrawing substituent (4-CN, **1g**), and halogen group (3,5-Cl_2 (**1t**), 4-Br (**1j**)) with 3-thienyl boronic acid and 3-furylboronic acid afforded the corresponding allenes in moderate to good yields with the addition of 2 equiv of water (Table 3).

Due to the importance of allenes with an axial chirality,[1,2i] we turned our attention to central-toaxial chirality transfer[18,19] for the synthesis of rather sensitive 1,3-diarylallenes since it has been observed that serious racemization occurred to optically active allene (*S*)-**3pj** under the optimized reaction conditions, although it is NOT reacemized in the absence of Pd(0) at 30 °C in dioxane (Scheme 4).



Scheme 4 The sensitivity of (*R*)-**3pj**

Thus, a rapid coupling before racemization starts is required. After extensively screening, we observed that the coupling reaction in the presence of 2 equiv of water completed within 30 minutes with high efficiency of chirality transfer (Scheme 5). The absolute configuration of (*S*)-**3qj** was established by single-crystal X-ray diffraction study (Figure 1).[20]



Scheme 5 Stereospecific synthesis of unstable diarylallenes

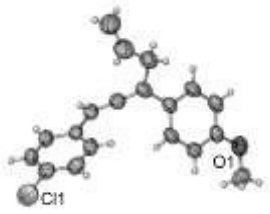


Figure 1 ORTEP representation of (*S*)-**3qj**

Ligands **L4**-**L7** were synthesized and explored to further reveal the role of formyl group in **L3** in this coupling reaction (Scheme 6). Benzylic alcohol **L4**[21] afforded only 19% of allene product **3aa**. The yield of **3aa** dropped to 20% when *p*-(diphenylphosphino)benzaldehyde (**L5**[22]) was used: **L3** and **L5** have a very similar electronic effect in the catalytic process. In addition, it should be noted that if the C=O is replaced with

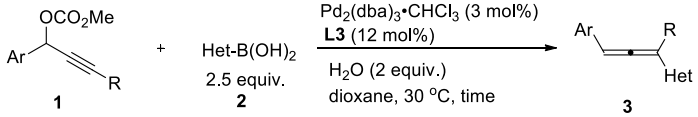
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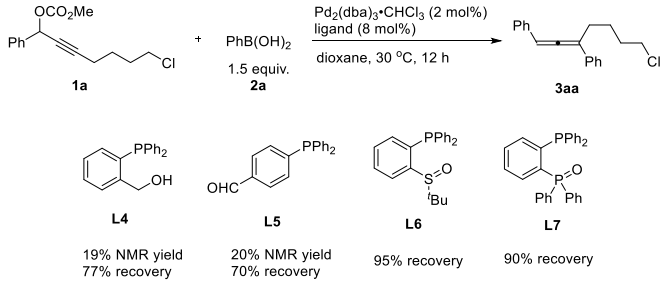
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Table 3 The reaction with heteroarylboronic acid[a]

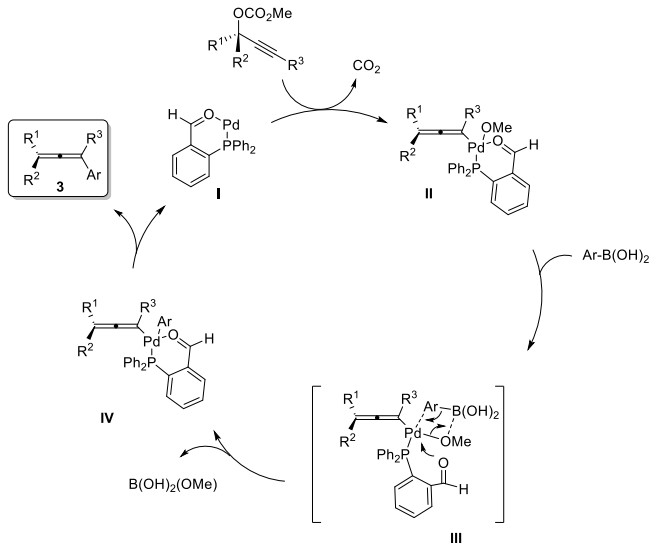


Entry	Ar	R	Het	Time (h)	Yield (%)[b]
1[c]	4-NCC_6H_4	^{n}Bu (**1g**)	3-thienyl (**2n**)	12	52 (**3gn**)[d]
2[c]	3,5-$Cl_2C_6H_3$	^{n}Bu (**1t**)	3-thienyl (**2n**)	12	61 (**3tn**)[e]
3[c]	4-BrC_6H_4	^{n}Bu (**1j**)	3-thienyl (**2n**)	12	72 (**3jn**)
4	4-BrC_6H_4	^{n}Bu (**1j**)	3-furyl (**2o**)	24	53 (**3jo**)[f]
5	Ph	$(CH_2)_4Cl$ (**1a**)	3-furyl (**2o**)	24	53 (**3ao**)[g]

[a]The reaction was conducted with 1 mmol of **1**, 2.5 mmol of **2**, 0.03 mmol of $Pd_2(dba)_3{\cdot}CHCl_3$, 0.12 mmol of **L3**, and 2 mmol of H_2O in 2 mL of dioxane for 12 h or 24 h at 30 °C under Ar atmosphere. [b]Isolated yield. [c]The reaction was conducted without water. [d]8% of **1g** were recovered. [e]3% of **1t** were recovered. [f]23% of **1j** were recovered. [g]17% of **1a** were recovered



Scheme 6 The unique role of formyl group



Scheme 7 Proposed mechanism-the role of the formyl entity

S=O or P=O, i.e., **L6**[23] and **L7**,[24] the reaction failed. A OMe group also failed to paly this role (see Scheme 3).

Based on these data, we believe that the overall catalytic cycle requires the help of the rather flexible coordination ability of the formyl group (Scheme 7):[25] with the hemilabile nature of this ligand, the coordination-unsaturated Pd(0)-**L3** complex **I** would undergo SN2' -type oxidative addition with propargylic carbonates in an *anti*-stereospecific manner to give the tetra-coordinated stereodefined allenylic palladium intermediate **II**. The bidentate ligand helps to keep this palladium intermediate **II** coordination-saturated, which avoids any possible racemization through the σ–π–σ isomerization process while still keeping the reactivity for the transmetalation. Subsequently, intermediate **II** underwent transmetalationwith arylboronic acids through the four-center transition states **III** by releasing the flexible coordination of the C=O unit. Water is facilating this step. Upon elimination of $B(OH)_2OMe$, the C=O bites back to form intermediate **IV**. Subsequent reductive elimination afforded the coupling product **3** and regenerates the catalytically active palladium(0) complex **I**.

Conclusions

In summary, we have developed a general and efficient catalyst system consisting of a hemilabile formyl-containing P,O-ligand, i.e., *o*-(diphenylphosphino)benzaldehyde, for the coupling reaction of propargylic carbonates with stable and commercially available arylboronic acids. The catalyst system functions under very mild conditions in absence of any base. A variety of useful functional groups are compatible with these reaction conditions to afford trisubstituted allenes, even tetrasubstituted allenes facilely. In addition, highly efficient central-to-axial chirality transfer is also realized with the help of water. Further studies on the details of the reaction mechanism as well as the asymmetric protocol are in progress in our laboratory.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21232006) is greatly appreciated. We thank Ms. Yujuan Cai in this group for reproducing the results of **3af**, **3ga**, and **3nl** presented in this study

Notes and references

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DOI: 10.1039/C6QO00490

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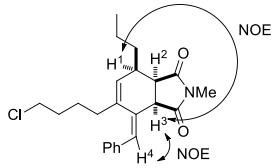
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